

Brutal Practice Paper B30 - Weather & Climate | Reproduction in Plants | Rational Numbers | Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. On Monday the maximum temperature was 34 degC and the minimum 18 degC; on Tuesday the maximum was 30 degC and the minimum 16 degC. By how many degrees did the daily temperature RANGE change from Monday to Tuesday? [\[Weather, Climate & Adaptations\]](#)

- (A) 2 degC
- (B) 0 degC
- (C) 4 degC
- (D) 6 degC
- (E) 16 degC
- (F) 30 degC
- (G) 14 degC

2. A sunflower head holds 800 seeds. If 65% of them germinate, how many seedlings grow? [\[Reproduction in Plants\]](#)

- (A) 520
- (B) 280
- (C) 480
- (D) 720
- (E) 260
- (F) 650

3. Find $-\frac{3}{4} + \frac{5}{6}$. [\[Rational Numbers\]](#)

- (A) $\frac{1}{12}$
- (B) $\frac{19}{12}$
- (C) $\frac{1}{2}$
- (D) $-\frac{1}{2}$
- (E) $-\frac{19}{12}$
- (F) $-\frac{1}{12}$

4. A rectangle 20 cm long and 14 cm wide has a path 2 cm wide running INSIDE, all along the boundary. What is the area of the path? [\[Perimeter & Area\]](#)

- (A) 160 cm²
- (B) 112 cm²
- (C) 120 cm²
- (D) 100 cm²
- (E) 140 cm²
- (F) 64 cm²

5. Rainfall over five days was 12 mm, 0 mm, 8 mm, 15 mm and 5 mm. What was the average daily rainfall over these five days? [\[Weather, Climate & Adaptations\]](#)
- (A) 40 mm
 - (B) 10 mm
 - (C) 8 mm
 - (D) 6 mm
 - (E) 7 mm
 - (F) 12 mm
 - (G) 9 mm
6. Yeast reproduces when a small bulge grows on the parent cell, enlarges and then separates as a new cell. This type of asexual reproduction is called: [\[Reproduction in Plants\]](#)
- (A) budding
 - (B) seed germination
 - (C) vegetative propagation
 - (D) fragmentation
 - (E) fertilisation
 - (F) pollination
7. Subtract $-\frac{2}{3}$ from $\frac{1}{6}$. [\[Rational Numbers\]](#)
- (A) $\frac{1}{2}$
 - (B) $-\frac{5}{6}$
 - (C) $\frac{5}{6}$
 - (D) $\frac{3}{2}$
 - (E) $-\frac{3}{6}$
 - (F) $\frac{1}{3}$
 - (G) $-\frac{1}{2}$
8. A circle is drawn touching all four sides of a square of side 28 cm. Using $\pi = \frac{22}{7}$, find the area inside the square but outside the circle. [\[Perimeter & Area\]](#)
- (A) 616 cm²
 - (B) 308 cm²
 - (C) 196 cm²
 - (D) 168 cm²
 - (E) 84 cm²
 - (F) 784 cm²
 - (G) 154 cm²
9. At a hill station the temperature was 5 degC at noon. It then fell by 3 degC each hour for the next 4 hours. What was the temperature after those 4 hours? [\[Weather, Climate & Adaptations\]](#)
- (A) 7 degC
 - (B) -3 degC
 - (C) -7 degC
 - (D) -12 degC
 - (E) -15 degC
 - (F) -8 degC
 - (G) 17 degC

- 10.** A complete flower has 5 sepals, 5 petals, 10 stamens and 1 pistil. If a plant bears 24 such flowers, how many stamens are there in all? [\[Reproduction in Plants\]](#)
- (A) 250
 - (B) 264
 - (C) 240
 - (D) 504
 - (E) 288
 - (F) 216
 - (G) 120
- 11.** Multiply $-\frac{5}{9}$ by $\frac{3}{10}$. [\[Rational Numbers\]](#)
- (A) $-\frac{1}{6}$
 - (B) $-\frac{15}{90}$
 - (C) $\frac{1}{3}$
 - (D) $-\frac{5}{30}$
 - (E) $\frac{1}{6}$
 - (F) $-\frac{8}{19}$
- 12.** A rectangular plot is 90 m by 50 m. Fencing costs Rs 30 per metre. What is the total cost of fencing it once around? [\[Perimeter & Area\]](#)
- (A) Rs 9000
 - (B) Rs 7000
 - (C) Rs 16800
 - (D) Rs 4500
 - (E) Rs 8400
 - (F) Rs 135000
- 13.** A town's annual rainfall used to be 1200 mm. This year it received 1500 mm. What is the percent increase in rainfall? [\[Weather, Climate & Adaptations\]](#)
- (A) 125%
 - (B) 15%
 - (C) 80%
 - (D) 25%
 - (E) 30%
 - (F) 300%
 - (G) 20%
- 14.** Coconuts can drift across the sea and sprout on distant shores. By which agent are their seeds mainly dispersed? [\[Reproduction in Plants\]](#)
- (A) gravity alone
 - (B) self-dispersal by the leaf
 - (C) explosion of the fruit
 - (D) animals
 - (E) wind
 - (F) insects only
 - (G) water

15. Divide $-7/12$ by $-14/15$. [\[Rational Numbers\]](#)

- (A) $-8/5$
- (B) $8/5$
- (C) $-5/8$
- (D) $5/8$
- (E) $98/180$
- (F) $5/16$

16. A square of side 18 cm and a rectangle have equal perimeters. The rectangle is 24 cm long. Find the area of the rectangle. [\[Perimeter & Area\]](#)

- (A) 288 cm²
- (B) 216 cm²
- (C) 432 cm²
- (D) 144 cm²
- (E) 240 cm²
- (F) 360 cm²
- (G) 324 cm²

17. The midday temperatures for a week were 28, 30, 29, 31, 32, 27 and 30 degC. Find the mean midday temperature, rounded to the nearest whole degree. [\[Weather, Climate & Adaptations\]](#)

- (A) 32 degC
- (B) 30 degC
- (C) 207 degC
- (D) 29 degC
- (E) 27 degC
- (F) 28 degC
- (G) 31 degC

18. A pea pod holds 8 peas. In a field of 40 pods, 75% of all the peas are healthy. How many healthy peas are there? [\[Reproduction in Plants\]](#)

- (A) 320
- (B) 80
- (C) 6
- (D) 240
- (E) 30
- (F) 120

19. What is the reciprocal of $-8/5$? [\[Rational Numbers\]](#)

- (A) 1
- (B) $8/-5$ is not defined
- (C) $8/5$
- (D) $-5/8$
- (E) $-8/5$
- (F) 0

20. A floor 6 m by 4.5 m is tiled with tiles measuring 30 cm by 25 cm. How many whole tiles are needed?

[Perimeter & Area]

- (A) 400
- (B) 300
- (C) 36
- (D) 720
- (E) 90
- (F) 360
- (G) 450

21. A maximum-minimum thermometer shows a maximum of 41 degC and a minimum of 26 degC for one day. The 'mean daily temperature' is taken as the average of these two. Find it. [Weather, Climate & Adaptations]

- (A) 15 degC
- (B) 32.5 degC
- (C) 33.5 degC
- (D) 34 degC
- (E) 33 degC
- (F) 67 degC

22. A maize cob has 12 rows of grains with 35 grains in each row. Since each grain is a seed, how many seeds are on the cob? [Reproduction in Plants]

- (A) 400
- (B) 240
- (C) 420
- (D) 47
- (E) 360
- (F) 384
- (G) 432

23. The additive inverse of $-\frac{11}{7}$ is: [Rational Numbers]

- (A) $-\frac{11}{7}$
- (B) $\frac{7}{11}$
- (C) $\frac{11}{7}$
- (D) 1
- (E) $-\frac{1}{7}$
- (F) $-\frac{7}{11}$
- (G) 0

24. A triangle has a base of 26 cm and a height of 15 cm. Find its area. [Perimeter & Area]

- (A) 180 cm²
- (B) 41 cm²
- (C) 169 cm²
- (D) 390 cm²
- (E) 260 cm²
- (F) 210 cm²
- (G) 195 cm²

- 25.** During a storm, 5 cm of rain fell on a flat terrace measuring 8 m by 5 m. What volume of water fell on it, in litres? ($1 \text{ m}^3 = 1000 \text{ litres}$) [\[Weather, Climate & Adaptations\]](#)
- (A) 4000 litres
 - (B) 2000 litres
 - (C) 40 litres
 - (D) 250 litres
 - (E) 200 litres
 - (F) 20000 litres
 - (G) 1000 litres
- 26.** A sunflower plant produces 1200 seeds. Birds eat $\frac{1}{4}$ of them, and then $\frac{1}{3}$ of the REMAINING seeds rot. How many seeds are left? [\[Reproduction in Plants\]](#)
- (A) 800
 - (B) 300
 - (C) 500
 - (D) 600
 - (E) 400
 - (F) 700
- 27.** Express the number $-\frac{18}{24}$ using its standard form, meaning lowest terms together with a positive denominator. What do you get? [\[Rational Numbers\]](#)
- (A) $\frac{3}{4}$
 - (B) $-\frac{3}{4}$
 - (C) $\frac{18}{24}$
 - (D) $-\frac{4}{3}$
 - (E) $\frac{4}{3}$
 - (F) $\frac{9}{12}$
- 28.** A parallelogram of base 16 cm and height 9 cm has the same area as a square. Find the side of the square. [\[Perimeter & Area\]](#)
- (A) 18 cm
 - (B) 24 cm
 - (C) 6 cm
 - (D) 36 cm
 - (E) 12 cm
 - (F) 25 cm
 - (G) 144 cm
- 29.** In a 30-day month it rained on 12 days. What percent of the days were rainy? [\[Weather, Climate & Adaptations\]](#)
- (A) 40%
 - (B) 60%
 - (C) 30%
 - (D) 12%
 - (E) 36%
 - (F) 48%
 - (G) 25%

- 30.** Transfer of pollen from the anther of one flower to the stigma of a flower on ANOTHER plant of the same kind is called: [\[Reproduction in Plants\]](#)
- (A) fragmentation
 - (B) germination
 - (C) budding
 - (D) double fertilisation
 - (E) cross-pollination
 - (F) fertilisation
- 31.** Which of these is the SMALLEST: $-\frac{1}{2}$, $-\frac{2}{3}$, $-\frac{3}{4}$, $-\frac{1}{4}$? [\[Rational Numbers\]](#)
- (A) $-\frac{3}{4}$
 - (B) $-\frac{2}{3}$
 - (C) $-\frac{1}{2}$
 - (D) 0
 - (E) $-\frac{1}{4}$
 - (F) they are all equal
 - (G) $-\frac{1}{3}$
- 32.** A wheel has radius 21 cm. Using $\pi = \frac{22}{7}$, how far does it travel in one full turn? [\[Perimeter & Area\]](#)
- (A) 154 cm
 - (B) 132 cm
 - (C) 66 cm
 - (D) 88 cm
 - (E) 1386 cm
 - (F) 42 cm
 - (G) 264 cm
- 33.** Which statement describes the CLIMATE of a place rather than today's weather? [\[Weather, Climate & Adaptations\]](#)
- (A) It is raining heavily right now.
 - (B) Strong winds are blowing this afternoon.
 - (C) Cherrapunji receives very heavy rainfall every year over many years.
 - (D) It is cloudy and humid this morning.
 - (E) The sky cleared up an hour ago.
 - (F) The temperature today is 31 degC.
 - (G) A thunderstorm is expected tonight.
- 34.** A farmer plants 5 rows of sugarcane using stem cuttings, with 48 cuttings per row. If 90% of the cuttings sprout, how many sprout? [\[Reproduction in Plants\]](#)
- (A) 210
 - (B) 192
 - (C) 24
 - (D) 216
 - (E) 200
 - (F) 240
 - (G) 250

35. Which of these rational numbers lies between $\frac{1}{4}$ and $\frac{1}{2}$? [\[Rational Numbers\]](#)

- (A) $\frac{2}{3}$
- (B) $\frac{1}{16}$
- (C) $\frac{5}{6}$
- (D) $\frac{1}{8}$
- (E) $\frac{3}{8}$
- (F) $\frac{5}{8}$

36. A circular park has radius 35 m. Using $\pi = \frac{22}{7}$, find its area. [\[Perimeter & Area\]](#)

- (A) 3850 m²
- (B) 1225 m²
- (C) 220 m²
- (D) 2450 m²
- (E) 7700 m²
- (F) 1925 m²
- (G) 110 m²

37. A polar bear has a thick layer of blubber (fat) and white fur. Which pair of needs do these features MOST directly serve? [\[Weather, Climate & Adaptations\]](#)

- (A) Seeing in the dark and hearing prey.
- (B) Flying long distances and avoiding rain.
- (C) Running fast and tolerating heat.
- (D) Breathing underwater and digging burrows.
- (E) Keeping warm and staying camouflaged in snow.
- (F) Swimming fast and finding food underground.

38. Bryophyllum (the sprout-leaf plant) can grow whole new plants from buds located on the: [\[Reproduction in Plants\]](#)

- (A) surface of its fruits
- (B) margins (edges) of its leaves
- (C) tips of its roots
- (D) centre of its seeds
- (E) inside of its flowers
- (F) tips of its thorns
- (G) underside of its bark

39. On the number line, between which two whole numbers does $-\frac{7}{3}$ lie? [\[Rational Numbers\]](#)

- (A) at exactly -7
- (B) between -2 and -1
- (C) between -3 and -2
- (D) between 2 and 3
- (E) between -8 and -7
- (F) between -7 and -6

40. An L-shaped figure is made from a 10 cm by 8 cm rectangle with a 4 cm by 3 cm rectangle cut away from one corner. Find its area. [\[Perimeter & Area\]](#)
- (A) 72 cm²
 - (B) 100 cm²
 - (C) 92 cm²
 - (D) 68 cm²
 - (E) 56 cm²
 - (F) 12 cm²
41. Over four months a place recorded rainfall of 30, 45, 60 and 25 mm. Express the wettest month's rainfall as a fraction of the total, in lowest terms. [\[Weather, Climate & Adaptations\]](#)
- (A) $\frac{2}{5}$
 - (B) $\frac{8}{3}$
 - (C) $\frac{3}{8}$
 - (D) $\frac{1}{3}$
 - (E) $\frac{3}{4}$
 - (F) $\frac{6}{16}$
42. Out of 60 flowers on a plant, 45 were pollinated and went on to form fruit. What fraction of the flowers formed fruit, in lowest terms? [\[Reproduction in Plants\]](#)
- (A) $\frac{1}{4}$
 - (B) $\frac{9}{12}$
 - (C) $\frac{3}{5}$
 - (D) $\frac{45}{60}$
 - (E) $\frac{4}{3}$
 - (F) $\frac{3}{4}$
 - (G) $\frac{15}{20}$
43. Simplify: $\frac{2}{3} + (-\frac{4}{5}) - (-\frac{1}{15})$. [\[Rational Numbers\]](#)
- (A) $\frac{1}{3}$
 - (B) $-\frac{1}{15}$
 - (C) $-\frac{7}{15}$
 - (D) $\frac{1}{15}$
 - (E) $\frac{7}{15}$
 - (F) $-\frac{1}{3}$
44. A path 1 m wide runs all around the OUTSIDE of a rectangular garden 12 m by 8 m. Find the area of the path. [\[Perimeter & Area\]](#)
- (A) 40 m²
 - (B) 96 m²
 - (C) 52 m²
 - (D) 48 m²
 - (E) 44 m²
 - (F) 36 m²

45. Temperatures at five polar stations were -12, -7, -15, -4 and -22 degC. What is the difference between the warmest and the coldest? [\[Weather, Climate & Adaptations\]](#)

- (A) 26 degC
- (B) 18 degC
- (C) 8 degC
- (D) -18 degC
- (E) 22 degC
- (F) 12 degC
- (G) 30 degC

46. After a pollen grain reaches the stigma, the male and female cells fuse. This fusion is called: [\[Reproduction in Plants\]](#)

- (A) germination
- (B) budding
- (C) pollination
- (D) fragmentation
- (E) dispersal
- (F) fertilisation
- (G) transpiration

47. Evaluate $(-2/3) \times (9/8) \times (-4/5)$. [\[Rational Numbers\]](#)

- (A) $-3/5$
- (B) $-6/5$
- (C) $6/5$
- (D) $-5/3$
- (E) $5/3$
- (F) $3/5$
- (G) $72/120$

48. A square field has side 50 m. A boy runs all the way around it 8 times. What total distance does he run? [\[Perimeter & Area\]](#)

- (A) 2500 m
- (B) 400 m
- (C) 20000 m
- (D) 1600 m
- (E) 200 m
- (F) 800 m

49. A desert animal must survive scorching days and water scarcity. Which adaptation BEST helps it conserve water? [\[Weather, Climate & Adaptations\]](#)

- (A) It huddles in groups to stay warm.
- (B) It has webbed feet for swimming.
- (C) It passes very little water as waste and tolerates large body-temperature swings.
- (D) It has gills to absorb moisture.
- (E) It hibernates underwater all summer.
- (F) It has a thick fat layer and white fur.
- (G) It grows broad leaves to catch rain.

50. After fertilisation, the OVARY of a flower develops into the: [\[Reproduction in Plants\]](#)

- (A) fruit
- (B) root
- (C) seed
- (D) another ovary
- (E) new flower
- (F) leaf

51. Find $(-3/7) \div (6/-14)$. [\[Rational Numbers\]](#)

- (A) 2
- (B) -2
- (C) 1
- (D) $7/3$
- (E) -1
- (F) $-1/2$
- (G) $1/2$

52. A wall 5 m long and 3 m high is to be painted at Rs 40 per square metre. Find the total cost. [\[Perimeter & Area\]](#)

- (A) Rs 60
- (B) Rs 200
- (C) Rs 320
- (D) Rs 600
- (E) Rs 640
- (F) Rs 1500
- (G) Rs 480

53. City A's average temperature is 8 degC and City B's is -6 degC. What is the average of the two cities' temperatures? [\[Weather, Climate & Adaptations\]](#)

- (A) -1 degC
- (B) 2 degC
- (C) -2 degC
- (D) 14 degC
- (E) 1 degC
- (F) -7 degC

54. A flower's ovary contains 6 ovules; 5 of them are fertilised and become seeds. If the plant produces 30 such fruits, how many seeds form in total? [\[Reproduction in Plants\]](#)

- (A) 120
- (B) 36
- (C) 165
- (D) 35
- (E) 150
- (F) 180
- (G) 155

55. A diver descends at $-5/2$ metres each minute (negative means downward). Where is she, relative to her start, after 3 minutes? [\[Rational Numbers\]](#)

- (A) $15/2$ m
- (B) $-5/6$ m
- (C) $-1/2$ m
- (D) $-3/2$ m
- (E) $-15/2$ m
- (F) $-5/2$ m

56. A triangle has an area of 84 cm^2 and a base of 14 cm. Find its height. [\[Perimeter & Area\]](#)

- (A) 12 cm
- (B) 6 cm
- (C) 70 cm
- (D) 9 cm
- (E) 24 cm
- (F) 14 cm

57. A rain gauge collected 36 mm of rain over 3 hours at a steady rate. At that same rate, how much rain would fall in a full 24-hour day? [\[Weather, Climate & Adaptations\]](#)

- (A) 864 mm
- (B) 12 mm
- (C) 108 mm
- (D) 96 mm
- (E) 144 mm
- (F) 72 mm
- (G) 288 mm

58. Spirogyra, a green alga, breaks into smaller pieces and each piece grows into a new alga. This kind of asexual reproduction is called: [\[Reproduction in Plants\]](#)

- (A) fragmentation
- (B) binary fission
- (C) vegetative grafting
- (D) fertilisation
- (E) spore formation
- (F) pollination

59. Find the reciprocal of $(1/2 + 1/3)$. [\[Rational Numbers\]](#)

- (A) 1
- (B) $2/5$
- (C) $1/5$
- (D) $-6/5$
- (E) $5/6$
- (F) $6/5$
- (G) 5

- 60.** The boundary of a certain square measures 64 cm in total length. Work out the area enclosed by this square. [\[Perimeter & Area\]](#)
- (A) 16 cm²
 - (B) 64 cm²
 - (C) 256 cm²
 - (D) 128 cm²
 - (E) 1024 cm²
 - (F) 32 cm²
- 61.** Emperor penguins crowd tightly into a huddle during polar winters. What does huddling MAINLY achieve? [\[Weather, Climate & Adaptations\]](#)
- (A) It lets them breathe underwater.
 - (B) It helps them swim faster together.
 - (C) It keeps their eggs cool.
 - (D) It attracts prey to the group.
 - (E) It speeds up digestion.
 - (F) It blocks sunlight from their eyes.
 - (G) It lowers the surface exposed to cold air, reducing heat loss.
- 62.** A flower that has BOTH stamens (male parts) and a pistil (female part) is described as: [\[Reproduction in Plants\]](#)
- (A) a sterile flower
 - (B) bisexual (a perfect flower)
 - (C) a female flower only
 - (D) unisexual
 - (E) a male flower only
 - (F) a spore-bearing flower
 - (G) an incomplete flower
- 63.** Which is larger: $\frac{7}{9}$ or $\frac{5}{7}$? [\[Rational Numbers\]](#)
- (A) $\frac{12}{16}$
 - (B) cannot be decided
 - (C) they are equal
 - (D) both are negative
 - (E) $\frac{5}{7}$
 - (F) $\frac{7}{9}$
 - (G) $\frac{2}{16}$
- 64.** A semicircle has radius 7 cm. Using $\pi = \frac{22}{7}$, find its perimeter (the curved edge plus the straight diameter). [\[Perimeter & Area\]](#)
- (A) 36 cm
 - (B) 44 cm
 - (C) 29 cm
 - (D) 14 cm
 - (E) 22 cm
 - (F) 32 cm
 - (G) 50 cm

65. A weather forecast was correct on 18 of 24 days. What percent of the forecasts were correct? [\[Weather, Climate & Adaptations\]](#)

- (A) 80%
- (B) 25%
- (C) 75%
- (D) 60%
- (E) 72%
- (F) 66%

66. A pond had 250 water-hyacinth plants. Because they reproduce so fast, a week later there were 400. What is the percent increase in the number of plants? [\[Reproduction in Plants\]](#)

- (A) 62.5%
- (B) 40%
- (C) 50%
- (D) 37.5%
- (E) 150%
- (F) 60%
- (G) 160%

67. The temperature changed by $-\frac{3}{4}$ degC every hour for 8 hours. What was the total change? [\[Rational Numbers\]](#)

- (A) -3 degC
- (B) -6 degC
- (C) $-\frac{2}{3}$ degC
- (D) -12 degC
- (E) $-\frac{32}{3}$ degC
- (F) 6 degC
- (G) $-\frac{3}{2}$ degC

68. A rectangle has length : breadth = 5 : 3 and a perimeter of 64 cm. Find its area. [\[Perimeter & Area\]](#)

- (A) 60 cm²
- (B) 32 cm²
- (C) 120 cm²
- (D) 64 cm²
- (E) 384 cm²
- (F) 240 cm²
- (G) 480 cm²

69. In a tropical rainforest the average temperature of the hottest and coldest months differ by only about 2 degC, while a desert can swing about 18 degC across the year. How many times larger is the desert's annual swing? [\[Weather, Climate & Adaptations\]](#)

- (A) 8 times
- (B) 9 times
- (C) 20 times
- (D) 36 times
- (E) 7 times
- (F) 2 times
- (G) 16 times

70. Seeds bearing hooks or stiff spines (such as those of Xanthium) that cling to the fur of passing animals are dispersed by: [\[Reproduction in Plants\]](#)
- (A) animals
 - (B) gravity alone
 - (C) self-pollination
 - (D) insects pollinating them
 - (E) wind
 - (F) explosion of the fruit
71. Which of these is EQUAL to $-\frac{4}{6}$? [\[Rational Numbers\]](#)
- (A) $-\frac{3}{2}$
 - (B) $-\frac{8}{3}$
 - (C) $-\frac{2}{3}$
 - (D) $\frac{2}{3}$
 - (E) $-\frac{1}{3}$
 - (F) $-\frac{6}{4}$
 - (G) $-\frac{4}{3}$
72. A rectangular park 60 m by 40 m has two roads, each 5 m wide, one running along the length and one along the breadth, crossing in the middle. Find the total area of the roads. [\[Perimeter & Area\]](#)
- (A) 525 m²
 - (B) 475 m²
 - (C) 300 m²
 - (D) 450 m²
 - (E) 200 m²
 - (F) 500 m²
 - (G) 550 m²
73. Which feature BEST adapts a monkey to life in a dense tropical rainforest? [\[Weather, Climate & Adaptations\]](#)
- (A) The ability to store water for weeks.
 - (B) Webbed feet for walking on ice.
 - (C) A thick layer of blubber under the skin.
 - (D) Long, strong limbs and a gripping tail for climbing trees.
 - (E) Wings for migrating across oceans.
 - (F) Very large ears to lose body heat in cold.
 - (G) White fur for camouflage in snow.
74. Bread mould and ferns can reproduce by releasing tiny, lightweight structures that spread through the air. These structures are called: [\[Reproduction in Plants\]](#)
- (A) buds
 - (B) fragments
 - (C) ovules
 - (D) seeds
 - (E) spores
 - (F) pollen grains
 - (G) fruits

75. Find $\frac{5}{6} - \frac{7}{8} + \frac{1}{4}$. [\[Rational Numbers\]](#)

- (A) $-\frac{1}{24}$
- (B) $\frac{5}{24}$
- (C) $\frac{1}{3}$
- (D) $-\frac{5}{24}$
- (E) $\frac{7}{24}$
- (F) $\frac{11}{24}$
- (G) $\frac{1}{24}$

76. A floor 4 m by 3 m is covered with square tiles of side 20 cm, each costing Rs 12. Find the total cost of the tiles. [\[Perimeter & Area\]](#)

- (A) Rs 3600
- (B) Rs 360
- (C) Rs 4800
- (D) Rs 1200
- (E) Rs 2400
- (F) Rs 300

77. The temperature rose steadily from -5°C at 6 am to 19°C at 2 pm. What was the average rise per hour? [\[Weather, Climate & Adaptations\]](#)

- (A) 2.5°C per hour
- (B) 12°C per hour
- (C) 8°C per hour
- (D) 14°C per hour
- (E) 24°C per hour
- (F) 3°C per hour
- (G) 1.75°C per hour

78. A seed packet contains 250 seeds. In the morning $\frac{2}{5}$ of them are sown and in the evening another $\frac{1}{10}$ are sown. How many seeds remain unsown? [\[Reproduction in Plants\]](#)

- (A) 25
- (B) 100
- (C) 130
- (D) 75
- (E) 150
- (F) 125

79. Divide $-\frac{9}{16}$ by 3. [\[Rational Numbers\]](#)

- (A) $\frac{3}{16}$
- (B) $-\frac{3}{16}$
- (C) $-\frac{9}{16}$
- (D) $-\frac{9}{48}$
- (E) $-\frac{1}{16}$
- (F) $-\frac{3}{8}$
- (G) $-\frac{27}{16}$

80. A circular track has a circumference of 264 m. Using $\pi = \frac{22}{7}$, find its radius. [\[Perimeter & Area\]](#)

- (A) 42 m
- (B) 84 m
- (C) 48 m
- (D) 21 m
- (E) 264 m
- (F) 132 m
- (G) 6 m

81. A hill town's temperature drops about 6 degC for every 1000 m gained in height. At sea level the temperature is 28 degC. What is the temperature at 2500 m? [\[Weather, Climate & Adaptations\]](#)

- (A) 10 degC
- (B) 12 degC
- (C) 13 degC
- (D) 43 degC
- (E) 18 degC
- (F) 15 degC

82. A pollen grain on the stigma grows a tube down through the style to reach the ovule. The main job of this pollen tube is to: [\[Reproduction in Plants\]](#)

- (A) bring water up from the roots
- (B) make food for the flower
- (C) colour the petals brightly
- (D) carry the male gametes (cells) down to the ovule
- (E) release seeds into the air
- (F) attract insects with scent
- (G) absorb sunlight for the ovary

83. A rational number lying between $-\frac{1}{2}$ and $-\frac{1}{4}$ is: [\[Rational Numbers\]](#)

- (A) -1
- (B) $-\frac{3}{4}$
- (C) $-\frac{7}{8}$
- (D) $-\frac{1}{8}$
- (E) 0
- (F) $-\frac{5}{8}$
- (G) $-\frac{3}{8}$

84. A right-angled triangle is exactly half of a rectangle 24 cm by 10 cm. Find the triangle's area. [\[Perimeter & Area\]](#)

- (A) 34 cm²
- (B) 120 cm²
- (C) 68 cm²
- (D) 60 cm²
- (E) 17 cm²
- (F) 48 cm²
- (G) 240 cm²

85. During the summer monsoon over India, moisture-laden winds blow mainly in which direction? [\[Weather, Climate & Adaptations\]](#)

[Climate & Adaptations\]](#)

- (A) Straight downward from the sky.
- (B) From the land towards the sea.
- (C) From the sea towards the land.
- (D) In no fixed direction at all.
- (E) Only from north to south.
- (F) From space towards the ground.
- (G) From colder land to warmer land.

86. A flower has an ovary with 9 ovules. After pollination, $\frac{2}{3}$ of the ovules are fertilised and become seeds. How many seeds does this fruit contain? [\[Reproduction in Plants\]](#)

[\[Reproduction in Plants\]](#)

- (A) 3
- (B) 6
- (C) 4
- (D) 18
- (E) 5
- (F) 9

87. A rod $\frac{7}{2}$ m long is cut into pieces each $\frac{7}{8}$ m long. How many whole pieces are obtained? [\[Rational Numbers\]](#)

[Numbers\]](#)

- (A) $\frac{49}{16}$
- (B) 7
- (C) 8
- (D) 4
- (E) $\frac{1}{4}$
- (F) 16

88. Square A has side 15 cm and Square B has side 9 cm. By how much is the area of A greater than the area of B? [\[Perimeter & Area\]](#)

[\[Perimeter & Area\]](#)

- (A) 81 cm²
- (B) 100 cm²
- (C) 144 cm²
- (D) 36 cm²
- (E) 6 cm²
- (F) 306 cm²
- (G) 24 cm²

89. Two readings were taken with a maximum-minimum thermometer: the highest was 8 degC above zero and the lowest was 11 degC below zero. What was the day's temperature range? [\[Weather, Climate & Adaptations\]](#)

[\[Weather, Climate & Adaptations\]](#)

- (A) 88 degC
- (B) 19 degC
- (C) 3 degC
- (D) -19 degC
- (E) 11 degC
- (F) -3 degC

90. Which one of these plants reproduces by SPORES, without ever forming seeds? [\[Reproduction in Plants\]](#)

- (A) mustard
- (B) apple
- (C) fern
- (D) maize
- (E) pea
- (F) mango
- (G) sunflower

91. What must be added to $-\frac{5}{9}$ to get 0 ? [\[Rational Numbers\]](#)

- (A) $-\frac{5}{9}$
- (B) 0
- (C) $-\frac{9}{5}$
- (D) $\frac{9}{5}$
- (E) $\frac{4}{9}$
- (F) $\frac{5}{9}$

92. A square of side 14 cm has a semicircle ($\pi = \frac{22}{7}$) drawn outwards on its top edge, the diameter equal to the side. Find the total area of the figure. [\[Perimeter & Area\]](#)

- (A) 273 cm²
- (B) 350 cm²
- (C) 77 cm²
- (D) 196 cm²
- (E) 245 cm²
- (F) 154 cm²
- (G) 308 cm²

93. A place gets 80% of its 2000 mm yearly rainfall during the four monsoon months. How much rain falls in the rest of the year? [\[Weather, Climate & Adaptations\]](#)

- (A) 200 mm
- (B) 800 mm
- (C) 1600 mm
- (D) 400 mm
- (E) 160 mm
- (F) 1200 mm
- (G) 500 mm

94. A grower takes 6 cuttings from each of 35 rose plants. If 80% of all the cuttings take root, how many rooted plants are obtained? [\[Reproduction in Plants\]](#)

- (A) 200
- (B) 168
- (C) 180
- (D) 150
- (E) 42
- (F) 35
- (G) 210

95. Arrange $-\frac{2}{5}$, $\frac{3}{10}$ and $-\frac{1}{2}$ in ascending order (smallest first). [\[Rational Numbers\]](#)

- (A) $-\frac{1}{2}$, $-\frac{2}{5}$, $\frac{3}{10}$
- (B) $-\frac{2}{5}$, $-\frac{1}{2}$, $\frac{3}{10}$
- (C) $\frac{3}{10}$, $-\frac{1}{2}$, $-\frac{2}{5}$
- (D) $\frac{3}{10}$, $-\frac{2}{5}$, $-\frac{1}{2}$
- (E) $-\frac{1}{2}$, $\frac{3}{10}$, $-\frac{2}{5}$
- (F) they are already in ascending order as given
- (G) $-\frac{2}{5}$, $\frac{3}{10}$, $-\frac{1}{2}$

96. A carpet measuring 250 cm by 200 cm costs Rs 150 per square metre. Find its total cost. [\[Perimeter & Area\]](#)

- (A) Rs 7500
- (B) Rs 500
- (C) Rs 375
- (D) Rs 75
- (E) Rs 750
- (F) Rs 1500

97. Over a year a station logged these seasonal mean temperatures: 12 degC, 30 degC, 27 degC and 15 degC. What is the yearly mean temperature? [\[Weather, Climate & Adaptations\]](#)

- (A) 20 degC
- (B) 42 degC
- (C) 84 degC
- (D) 21 degC
- (E) 18 degC
- (F) 22 degC
- (G) 24 degC

98. A ripe balsam (touch-me-not) fruit suddenly bursts open and flings its seeds far away. This scattering is a form of seed dispersal by: [\[Reproduction in Plants\]](#)

- (A) gravity alone
- (B) the pollen tube
- (C) wind
- (D) the bursting (explosion) of the fruit itself
- (E) clinging to animals
- (F) floating insects
- (G) water

99. A tank is $\frac{3}{4}$ full. Then $\frac{1}{3}$ of the water that is in it is used. What fraction of the FULL tank is left? [\[Rational Numbers\]](#)

- (A) $\frac{3}{4}$
- (B) $\frac{5}{12}$
- (C) $\frac{1}{3}$
- (D) $\frac{7}{12}$
- (E) $\frac{1}{2}$
- (F) $\frac{2}{3}$
- (G) $\frac{1}{4}$

100. A parallelogram has sides 12 cm and 8 cm. The height drawn to the 12 cm side is 6 cm. Find the height drawn to the 8 cm side. [\[Perimeter & Area\]](#)

- (A) 8 cm
- (B) 12 cm
- (C) 16 cm
- (D) 3 cm
- (E) 4 cm
- (F) 9 cm
- (G) 6 cm